

TECHNICAL REPORT

**Thermal and Mass Optimization of Cryogenic Liquid
Hydrogen Storage for Earth–Mars Transit with
Application to Long-Duration Boil-off Mitigation**

*A Parametric Thermal Analysis and System-Level Mass Trade Study
Incorporating Time-Varying Solar Conditions*

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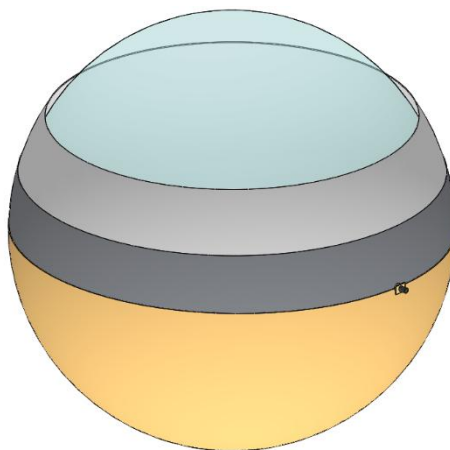


Figure 14: Cross-sectional view of optimal LH₂ tank thermal architecture.

Abstract

Cryogenic propellant storage for long-duration Mars missions is fundamentally challenged by continuous boil-off driven by radiative heat leak, which can consume a substantial fraction of the propellant load over a 240-day transit. This study develops a parametric thermal model of a spherical liquid hydrogen tank to evaluate passive multi-layer insulation, active cryocooler performance, and their combined effect on total system mass under a time-varying solar environment. A one-dimensional heat transfer model is developed based on the Stefan-Boltzmann law and MLI radiation theory. The external surface temperature is determined from a radiative equilibrium model that varies with heliocentric distance as the spacecraft travels from Earth to Mars. The optimal design is found at 11 MLI layers with 110 W of active cooling, yielding a total system mass of 348 kg and reducing boil-off from 5,241 kg to 583 kg, a saving of 4,658 kg of propellant. Constant-temperature assumptions underestimate cumulative boil-off by a factor of two, confirming that mission-phase thermal variation cannot be ignored in propellant margin calculations. Active cooling, despite its electrical power penalty, delivers substantial mass savings when traded against insulation and power system mass, making it a viable strategy for deep space cryogenic storage design.

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1.Introduction

The exploration of Mars, whether by robotic precursors or future human crews, places unprecedented demands on in-space propulsion systems. Among chemical propellants, liquid hydrogen (LH₂) and liquid oxygen (LOX) deliver the highest specific impulse, making them the preferred choice for deep space missions where every kilogram of propellant directly impacts payload capacity and mission feasibility. However, the extreme cryogenic temperatures required to maintain these propellants, 20.3 K for LH₂ and 90 K for LOX, introduce substantial engineering challenges. In the vacuum of space, radiative heat transfer becomes the dominant mechanism of thermal exchange, and even small heat leaks into cryogenic tanks drive continuous propellant vaporization, a process known as boil-off, that accumulates significantly over the duration of a long transit.

For a 6- to 9- month Earth–Mars transit, the cumulative effect of boil-off is not merely a parasitic loss but a critical design driver. Without mitigation, a significant fraction of the propellant load can be lost before arrival, undermining mission margins or requiring prohibitively large initial propellant masses. Two principal strategies exist to manage this heat load: passive insulation using multi-layer insulation (MLI), which reduces the incoming heat flux, and active cooling via cryocoolers, which removes heat from the tank. Neither strategy is without cost; MLI adds mass directly, while cryocoolers impose electrical power demands that translate into additional solar array mass. The problem therefore becomes one of optimization: finding the combination of insulation thickness and cooling capacity that minimizes total system mass for a given mission profile.

This study develops a parametric thermal model of a spherical LH₂ storage tank for a 240-day Earth–Mars transit. The model sweeps MLI layer count and cryocooler capacity across realistic design ranges under a time-varying solar environment, and one- and two-dimensional parametric sweeps are used to identify the optimal combination that minimizes total system mass. The results quantify the inadequacy of constant-temperature assumptions commonly used in early-phase analyses and demonstrate the mass savings enabled by active cooling under realistic mission conditions.

Section 2 presents the theoretical background on radiative heat transfer, MLI, and cryocooler performance, along with the solar environment model. Section 3 describes the numerical modeling approach and key assumptions used throughout the study. Section 4 presents the parametric results across passive-only performance, active cooling comparisons, and total mass optimization. Section 5 discusses the implications of these findings for mission design and identifies key sensitivities, and Section 6 states the conclusions and outlines potential avenues for further refinement.

2. Background and Theory

2.1 Cryogenic Propellant Storage Challenges

Liquid hydrogen (LH₂) is one of the coldest liquids used in engineering, with a boiling point of 20.3 K at one atmosphere [1]. Storing LH₂ for long-duration space missions is fundamentally challenging because any heat leak into the tank causes vaporization, a process known as boil-off. In the vacuum of space, convection is negligible, making radiation the dominant mode of heat transfer. As boil-off occurs, the tank pressure rises; unless the vapor is vented, structural integrity becomes a concern. Venting, however, results in direct propellant mass loss. Over a 240-day Earth–Mars transit, even a small steady heat leak accumulates into a significant loss of propellant, threatening mission feasibility. Addressing these challenges requires both passive insulation in the form of multi-layer insulation (MLI) and active cooling via cryocoolers, each targeting the heat leak from a different angle.

2.2 Radiative Heat Transfer and MLI

In the vacuum of space, thermal radiation is the only significant mode of heat transfer between the tank and its surroundings. For two concentric, gray surfaces at temperatures T_{hot} and T_{cold} , the net radiative heat flux is given by the Stefan-Boltzmann law:

$$Q = \varepsilon \sigma A (T_{hot}^4 - T_{cold}^4) \quad (1)$$

where Q is the heat transfer rate (W), ε is the emissivity of the surfaces (assumed equal), $\sigma = 5.67 \times 10^{-8} \text{ W/m}^2 \cdot \text{K}^4$ is the Stefan-Boltzmann constant, and A is the area (m²). Multi-layer insulation reduces this heat flux by inserting N thin radiation shields between the hot and cold boundaries. Each shield intercepts and re-radiates thermal energy, effectively adding thermal resistance in series. Assuming all shields have the same emissivity ε on both sides, the overall heat flux becomes:

$$Q = \frac{\sigma A (T_{hot}^4 - T_{cold}^4)}{(N + 1) \left(\frac{2}{\varepsilon} - 1 \right)} \quad (2)$$

The denominator captures the combined effect of multiple shields and their surface properties. The factor $(N+1)$ reflects that the total number of radiative gaps is one more than the number of shields; more layers increase thermal resistance, reducing heat flux. The term $(2/\varepsilon - 1)$ accounts for the resistance per gap: lower emissivity yields a larger denominator and thus lower heat transfer. The fourth-power dependence on temperature means that even modest increases in the hot boundary temperature cause disproportionately large increases in heat load, which is critical for mission phases near the Sun. In spacecraft applications, MLI typically uses aluminized mylar with an effective emissivity $\varepsilon \approx 0.03$ [2], providing high thermal resistance with minimal mass.

2.3 Boil-off Rate

Once the heat flux Q entering the cryogenic tank is known, the instantaneous boil-off rate $\dot{m}$ follows directly from energy conservation at the liquid-vapor interface. The tank is

assumed to remain at constant temperature (20.3 K at 1 atm) while liquid vaporizes, so all incoming heat is consumed by the latent heat of vaporization:

$$\dot{m} = \frac{Q}{L_v} \quad (3)$$

where $L_v = 446,000$ J/kg is the latent heat of vaporization of liquid hydrogen [1]. This linear relationship allows total propellant loss over a mission to be computed by numerically integrating the time-varying heat load.

2.4 Active Cooling and Coefficient of Performance

Active cooling employs a cryocooler, a thermodynamic device that removes heat from the tank by doing mechanical work. The net heat load on the propellant becomes:

$$Q_{net} = Q_{passive} - Q_{cool} \quad (4)$$

where Q_{cool} is the cooling power provided by the cryocooler. When Q_{cool} equals or exceeds the passive heat leak, zero boil-off can be achieved. The efficiency of the cryocooler is characterized by the coefficient of performance (COP), defined as:

$$\text{COP} = \frac{Q_{cool}}{P_{input}} \quad (5)$$

where P_{input} is the electrical power required. For cryogenic temperatures, the COP is inherently low because the second law of thermodynamics limits the maximum possible efficiency. The Carnot COP for a refrigerator operating between T_{cold} (20 K) and T_{hot} (300 K) is:

$$\text{COP}_{carnot} = \frac{T_{cold}}{T_{hot} - T_{cold}} = \frac{20}{300 - 20} \approx 0.07 \quad (6)$$

Practical cryocoolers typically achieve 15-30% of Carnot efficiency, giving actual COP values in the range 0.01-0.05 for systems operating near 20K. Even modest cooling at 20K therefore demands substantial electrical power, directly driving solar array size and overall spacecraft mass.

2.5 Total Mass Optimization

The design of a cryogenic storage system involves balancing competing mass contributions. The total system mass is expressed as:

$$M_{total} = M_{boiloff} + M_{MLI} + M_{power} \quad (7)$$

Here, $M_{boiloff}$ is the mass of propellant lost to boil-off over the mission, M_{MLI} is the mass of the multi-layer insulation, and M_{power} is the mass of the electrical power system needed to run the cryocooler. These terms trade off against one another. Adding more MLI layers reduces $M_{boiloff}$ but increases M_{MLI} . Increasing the cooling power reduces $M_{boiloff}$ but raises M_{power} because the power system must be larger to supply the required electrical energy. The optimal design point is where the sum of all three components is minimized,

balancing propellant savings against the mass cost of insulation and power hardware. In this study, M_{MLI} is calculated from the areal density of the MLI (taken as 0.3 kg/m² per layer, a typical value for flight-qualified MLI systems [2]) and the tank surface area. M_{power} is derived from the specific mass of the power system, which for solar arrays at Mars distance is approximately 0.083 kg/W, reflecting the reduced solar flux at Mars (590 W/m²) compared to Earth (1361 W/m² [3]), requiring a larger panel area to produce equivalent power output.

2.6 Solar Environment Model

The external thermal environment during the Earth–Mars transit varies with distance from the Sun. Solar flux follows the inverse square law:

$$S(r) = \frac{S_0}{r^2} \quad (8)$$

where $S_0 = 1361 \text{ W/m}^2$ is the solar constant at 1 AU [3] and r is the distance from the Sun in astronomical units (AU). The hot boundary temperature of the spacecraft's outer surface is determined by radiative equilibrium: absorbed solar radiation equals emitted thermal radiation. Assuming the surface is a gray body with solar absorptivity α and infrared emissivity ε_{outer} , the equilibrium temperature is:

$$T_{hot} = \left(\frac{\alpha S(r)}{\varepsilon_{outer} \sigma} \right)^{0.25} \quad (9)$$

For this study, the outer surface is assumed to be aluminized mylar, with $\alpha = 0.1$ and $\varepsilon_{outer} = 0.05$ [2]. These values produce a hot boundary temperature that decreases from approximately 467 K at Earth departure to 379 K at Mars arrival. This temperature variation directly drives the time-varying heat load evaluated throughout the mission.

3. Methodology

3.1 Tank geometry and parameters

The cryogenic storage system is modeled as a spherical liquid hydrogen (LH₂) tank with a radius of 2 m, chosen because a sphere minimizes surface area for a given volume, reducing radiative heat ingress per unit of propellant stored. The tank is assumed to operate at the normal boiling temperature of LH₂, 20.3 K at approximately 1 atm. Key physical and thermal parameters are listed in Table 1.

Parameter	Symbol	Value
Tank radius	r	2 m
LH ₂ boiling temperature	T_{cold}	20.3 K
Latent heat of vaporization	L_v	446,000 J/kg [1]

Parameter	Symbol	Value
Stefan–Boltzmann constant	σ	$5.67 \times 10^{-8} \text{ W/m}^2\text{K}^4$
MLI effective emissivity	ϵ	0.03
Outer surface absorptivity	α	0.1 [2]
Outer surface emissivity	ϵ_{outer}	0.05 [2]
Solar flux at 1 AU	S_0	1361 W/m^2 [3]

These parameters remain fixed throughout the analysis unless explicitly varied as part of a parametric sweep.

3.2 Numerical model and simulation setup

The thermal and mass models are implemented in Python using a time-discrete numerical framework over a 240-day Earth–Mars transit. The LH₂ tank is represented as a lumped-parameter control volume, and radiative heat transfer, boil-off, and active cooling power are evaluated at each time step and integrated over the mission duration to obtain cumulative propellant loss and total system mass. A fixed time step of 1 day is used, which provides sufficient temporal resolution to capture the gradual variation in solar flux between 1 AU and 1.524 AU while keeping computational cost low.

For each time step, the heliocentric distance $r(t)$ is updated assuming a linear interpolation between Earth and Mars orbital radii, the corresponding solar flux is computed from Equation (8)

$$S(t) = \frac{S_0}{r(t)^2}$$

with $S_0 = 1361 \text{ W/m}^2$ at 1 AU. The outer surface temperature is then evaluated using the radiative equilibrium relation given in Equation (9)

$$T_{\text{hot}}(t) = \left(\frac{\alpha S(t)}{\epsilon_{\text{outer}} \sigma} \right)^{1/4}$$

and this temperature is applied as the boundary condition to the MLI radiation model. At each step, the passive heat leak $Q_{\text{passive}}(t)$, optional cooling power Q_{cool} , net heat load $Q_{\text{net}}(t)$, and instantaneous boil-off rate $\dot{m}(t) = Q_{\text{net}}(t)/L_v$ (with $\dot{m}(t) \geq 0$) are computed, and the mass loss is accumulated over time using explicit time integration.

To map the full design space, the simulation is repeated automatically across a grid of MLI layer counts and cooling power values. Parametric sweeps are automated by nesting the time-marching solver within loops over MLI layer count N and cryocooler capacity Q_{cool} . For each (N, Q_{cool}) pair, the simulation outputs total boil-off mass, time history of boil-off rate, required electrical power $P_{\text{input}} = Q_{\text{cool}}/\text{COP}$, and derived power system mass using the assumed specific mass of 0.083 kg/W representative of solar array technology at Mars distance [3]. These results are post-processed to generate one- and two-dimensional trade-study plots,

including boil-off versus MLI layers, residual boil-off versus cooling power, input power versus cooling power, and total system mass contours over the (N, Q_{cool}) design space.

3.3 Passive thermal model

The passive thermal performance is modeled using a multi-layer insulation (MLI) radiation heat transfer formulation derived in section 2.2. The net radiative heat leak through the MLI blanket is given by Equation (2).

$$Q = \frac{\sigma A (T_{\text{hot}}^4 - T_{\text{cold}}^4)}{(N + 1) \left(\frac{2}{\varepsilon} - 1 \right)}$$

where A is the tank surface area, T_{hot} is the outer-surface temperature, T_{cold} is the propellant temperature, ε is the effective emissivity, and N is the number of MLI layers. The instantaneous boil-off rate follows from Equation (3).

$$\dot{m} = \frac{Q}{L_v}$$

A parametric sweep of N from 1 to 100 layers is performed to map insulation performance and identify the point of diminishing returns, beyond which additional layers provide negligible thermal benefit.

3.4 Active cooling model

An active cryocooler model is introduced to reduce the net heat influx to the LH₂ tank. The net heat load is given by

$$Q_{\text{net}} = Q_{\text{passive}} - Q_{\text{cool}}$$

where Q_{passive} is the radiative heat leak predicted by the MLI model and Q_{cool} is the cooling capacity of the cryocooler. The corresponding electrical power requirement is modeled using a coefficient of performance (COP):

$$P_{\text{input}} = \frac{Q_{\text{cool}}}{\text{COP}}$$

A COP range of 0.01–0.05 is examined to represent state-of-the-art space cryocoolers [4]. Zero boil-off (ZBO) is defined as the condition where $Q_{\text{cool}} \approx Q_{\text{passive}}$, resulting in $\dot{m} \approx 0$ over the mission duration.

3.5 Total mass optimization

The total system mass is defined as

$$M_{\text{total}} = M_{\text{boil-off}} + M_{\text{MLI}} + M_{\text{power}}$$

where $M_{\text{boil-off}}$ is the cumulative propellant mass lost to boil-off, M_{MLI} is the mass of the insulation system, and M_{power} is the mass of the power generation hardware required to drive the cryocooler. The MLI mass is calculated from an areal density of 0.3 kg/m^2 per layer applied to the spherical surface, and the power system mass is computed using a specific mass of 0.083 kg/W representative of solar power at Mars distance [3]. A two-dimensional parametric sweep over MLI layer count N and cooling capacity Q_{cool} is carried out to map the total mass landscape and to identify the configuration that minimizes M_{total} while satisfying ZBO or reduced-boil-off objectives.

3.6 Mission phase thermal model

The external thermal environment is modeled as a function of heliocentric distance to capture the changing solar flux along the Earth–Mars transfer. The incident solar flux is approximated as

$$S(t) = \frac{S_0}{r(t)^2}$$

where $r(t)$ is the spacecraft distance from the Sun in astronomical units and $S_0 = 1361 \text{ W/m}^2$ is the solar constant at 1 AU [3]. Assuming radiative equilibrium for the outer surface, the effective hot-side temperature is given by

$$T_{\text{hot}}(t) = \left(\frac{\alpha S(t)}{\epsilon_{\text{outer}} \sigma} \right)^{\frac{1}{4}}$$

which produces a time-varying T_{hot} decreasing from approximately 467 K near 1 AU to 379 K at 1.524 AU over the 240-day transit. This temperature profile serves as the time-varying boundary condition for both the passive and active thermal models throughout the 240-day transit.

3.7 Assumptions

Several simplifying assumptions are made to keep the model manageable while preserving the dominant physics. These are summarized in Table 2.

Assumption	Description
Vacuum environment	Convection heat transfer is neglected
Uniform tank temperature	No internal thermal gradients in LH_2
Ideal MLI behavior	Spacer conduction and edge effects neglected
Constant properties	ϵ and α independent of temperature
Linear trajectory	Hohmann transfer dynamics not modeled

Assumption	Description
Single propellant	Only LH ₂ is considered
Spin-stabilization neglected	Provides conservative maximum solar heat input estimate

Together these assumptions allow the study to focus on the thermal and mass trade-offs between insulation and active cooling, without introducing complexity that would obscure the primary design drivers.

3.8 Tank Design Overview

The physical configuration of the optimal tank design is illustrated in Figures 10–13. The assembly consists of a 2m radius spherical LH₂ tank enclosed by an 11-layer MLI blanket and an outer structural shell. A cryocooler cold finger penetrates the outer shell and MLI blanket to maintain thermal contact with the inner tank wall. The cross-sectional views show the distinct layer architecture that forms the basis of the thermal model described in the preceding sections.

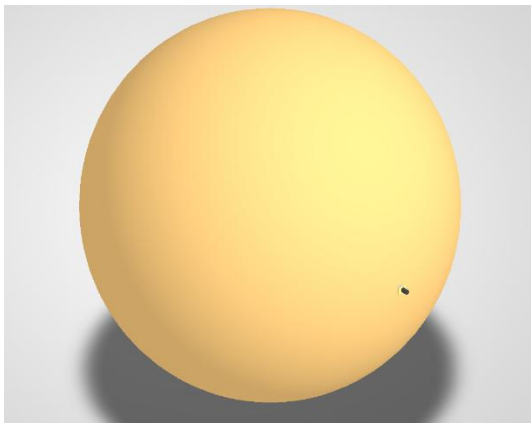


Figure 10

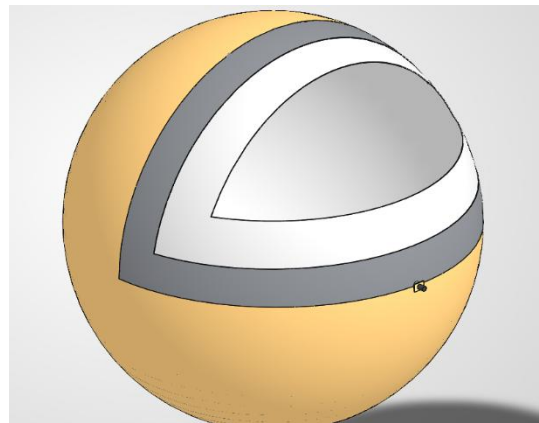


Figure 11

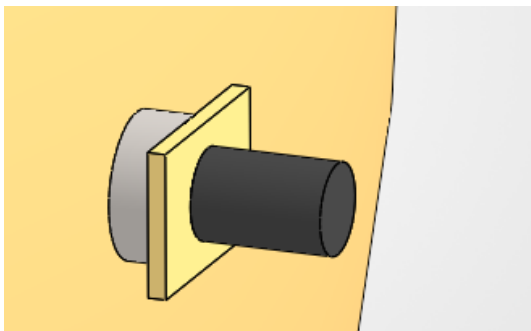


Figure 12

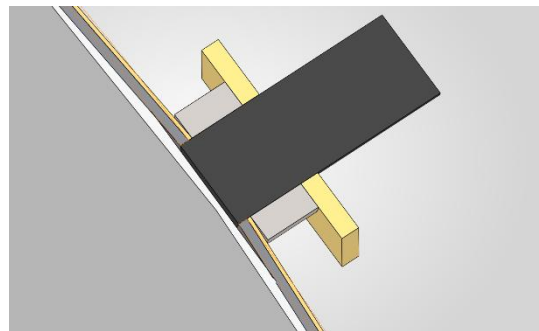


Figure 13

Figure 10: Exterior view of optimal cryogenic LH₂ storage tank assembly: 2m sphere with MLI blanket and cryocooler penetration

Figure 11: Cross-sectional view of the tank assembly showing inner LH₂ tank, MLI blanket, and outer structural shell

Figure 12: Cryocooler cold finger penetration assembly — collar, bracket, and finger components

Figure 13: Wall layer stack detail showing the inner tank wall, MLI blanket, and outer shell as distinct thermal barriers

4. Results

The results are presented in four parts: passive-only performance, active cooling comparisons, total mass optimization, and mission-phase analysis.

4.1 Passive Model

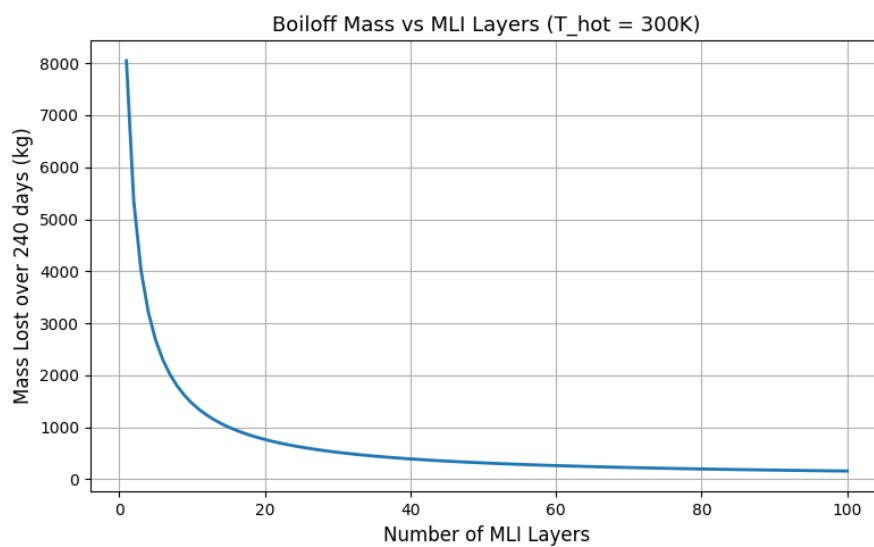


Figure 1: Boil-off mass loss vs number of MLI layers over 240-day transit ($T_{hot} = 300K$, passive only)

The passive thermal performance of the LH₂ storage system was first characterized using the MLI-only configuration. **Figure 1** presents the total mass of propellant lost due to boil-off over the 240-day Earth–Mars transit as a function of the number of MLI layers, with a fixed external temperature of 300 K. The curve exhibits a hyperbolic decay, where the boil-off mass decreases rapidly with the first 20–30 layers, after which further additions yield progressively smaller reductions. This behavior is governed by the inverse relationship between the net heat flux and the number of layers, where each additional layer adds thermal resistance, with the rate of improvement diminishing as N increases. Quantitatively, increasing the layer count from 5 to 30 reduces the boil-off mass from 12,850 kg to 2,680 kg, a reduction of nearly 80%, while adding layers beyond 30 to 100 layers only lowers the boil-off further to approximately 1,140 kg.

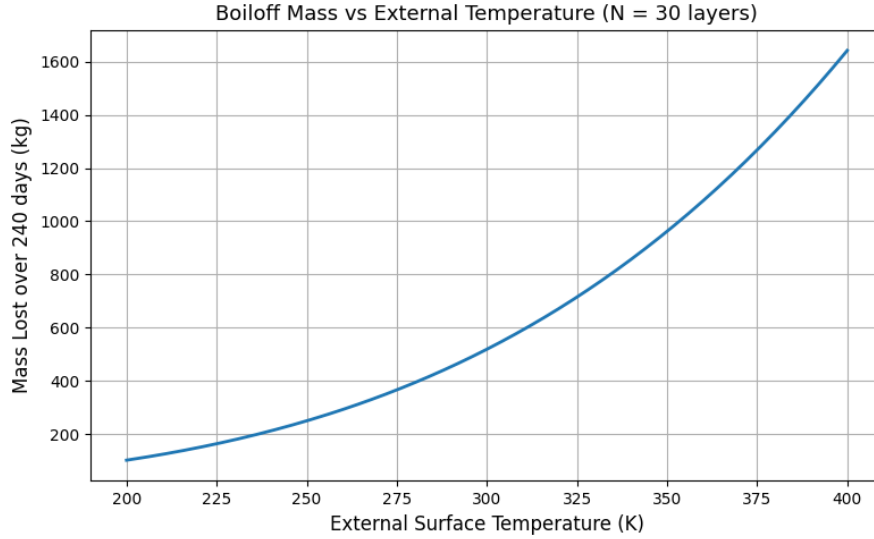


Figure 2: Boil-off mass loss vs external surface temperature over 240-day transit ($N = 30$ layers, passive only)

Figure 2 illustrates the sensitivity of the passive system to the external environment, showing boil-off mass over 240 days as a function of the hot boundary temperature for a fixed MLI layer count of 30. The trend follows a power-law relationship consistent with the Stefan-Boltzmann law, where the heat load, and therefore the boil-off, scales with T_{hot}^4 . The strong non-linearity highlights the critical impact of the early mission phase when the spacecraft is near Earth and exposed to higher solar flux. For example, increasing the external temperature from 200 K to 400 K results in a twelve-fold increase in propellant loss, rising from 260 kg to over 3,100 kg. A single representative temperature is therefore insufficient for accurate propellant loss estimation; the full thermal trajectory must be modeled.

4.2 Active Cooling Comparison

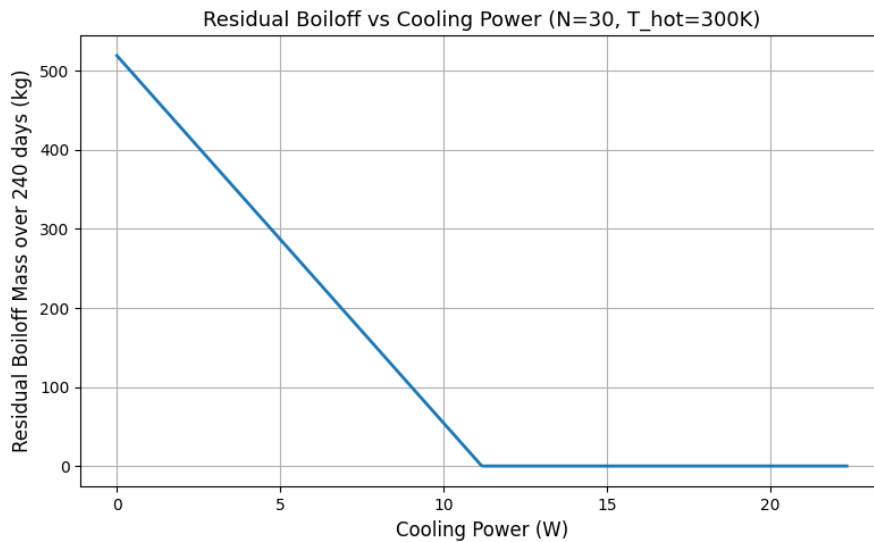


Figure 3: Residual boil-off mass vs cryocooler cooling power ($N = 30$, $T_{hot} = 300K$)

The effectiveness of integrating a cryocooler was evaluated by fixing the passive insulation at 30 layers and a hot boundary of 300 K, then varying the active cooling power. **Figure 3** plots

the residual boil-off mass over the 240-day transit as a function of the cooling power Q_{cool} . The relationship is linear, with boil-off decreasing proportionally to the heat removed until reaching a zero-boil-off threshold at $Q_{cool} \approx 11$ W. Beyond this point, the cryocooler extracts heat faster than the passive heat leak, eliminating propellant loss entirely; the curve flattens because no further reduction in boil-off is possible once the tank is maintained below the boiling point.

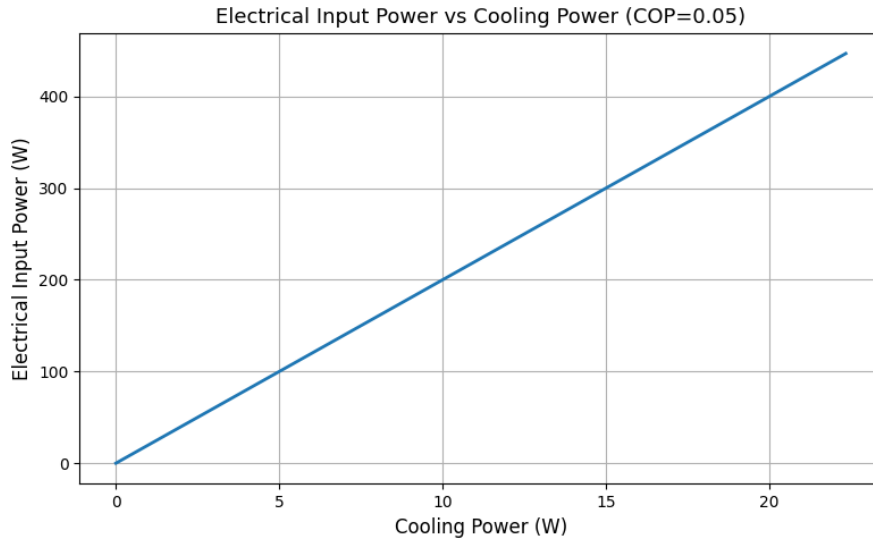


Figure 4: Electrical input power vs cryocooler cooling power ($COP = 0.05$)

The electrical power required to drive the cryocooler is shown in **Figure 4** as a function of Q_{cool} . The linear trend follows directly from $P_{input} = Q_{cool}/COP$, where a constant coefficient of performance (COP) of 0.05 was assumed based on representative cryocooler efficiency at 20 K [4]. Achieving zero boil-off at the 11 W threshold therefore demands approximately 220 W of electrical power. This value is significant in the context of a Mars transit, where solar panel output is constrained by the reduced solar flux at Mars distance (590 W/m^2 [3]) and the system must supply continuous power throughout the journey.

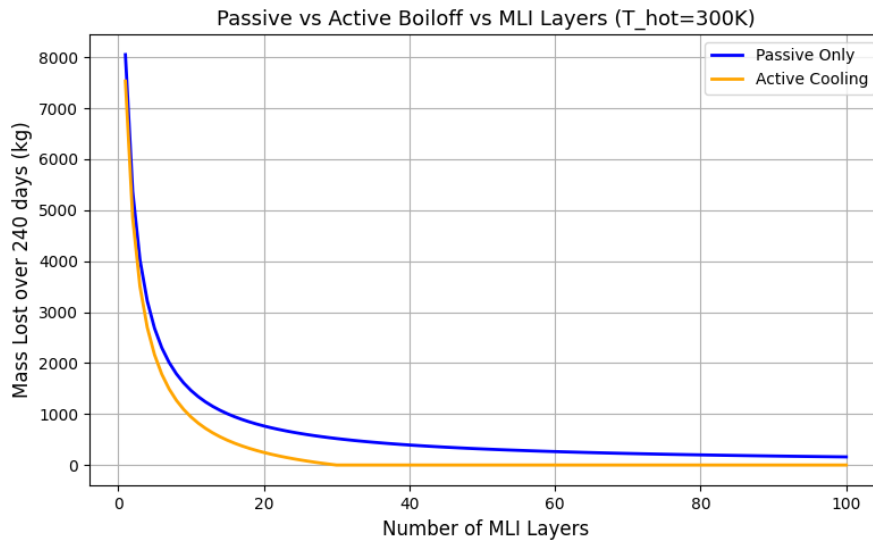


Figure 5: Passive vs active boil-off mass vs MLI layer count ($T_{hot} = 300K$, $Q_{cool} = Q_{passive}$ at $N=30$)

A direct comparison between passive and active configurations across a range of MLI layer counts is provided in **Figure 5**, where both curves show the total boil-off mass over 240 days. The passive curve decays hyperbolically as layers increase, while the active curve remains at zero boil-off for all layer counts above approximately 10 layers when Q_{cool} is fixed at 110 W. Below this threshold, the active system cannot fully compensate for the high passive heat leak, leading to residual boil-off. At the optimal passive-only design of 11 layers, the active configuration eliminates boil-off entirely. This confirms that active cooling can compensate for reduced insulation when mass or packaging constraints limit MLI layer count.

4.3 Total Mass Optimization

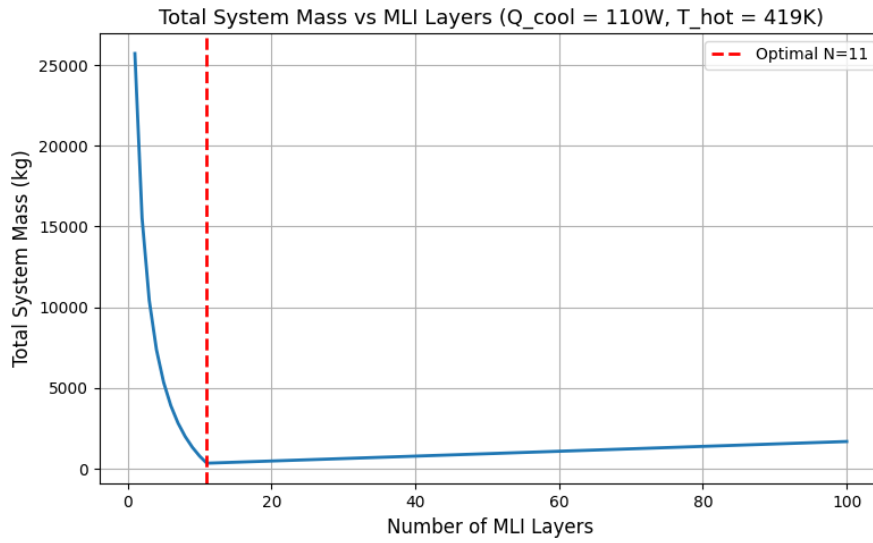


Figure 6: Total system mass vs MLI layer count ($Q_{cool} = 110W$, $T_{hot} = 419K$)

The trade-off between boil-off mass, MLI mass, and power system mass was evaluated to identify the minimum total system mass. **Figure 6** shows total system mass as a function of the number of MLI layers for a fixed cooling power of 110 W, corresponding to the design point chosen after convergence with the time-varying thermal model. The curve forms a distinct V-shape: on the left side, boil-off mass dominates due to inadequate insulation, while on the right side, the increasing mass of additional MLI layers outweighs the diminishing boil-off savings. The minimum occurs at 11 layers, with a total mass of 350 kg. This minimum exists because the competing mass penalties, propellant loss versus insulation mass, balance at this point, consistent with typical optimization of multi-layer insulation systems.

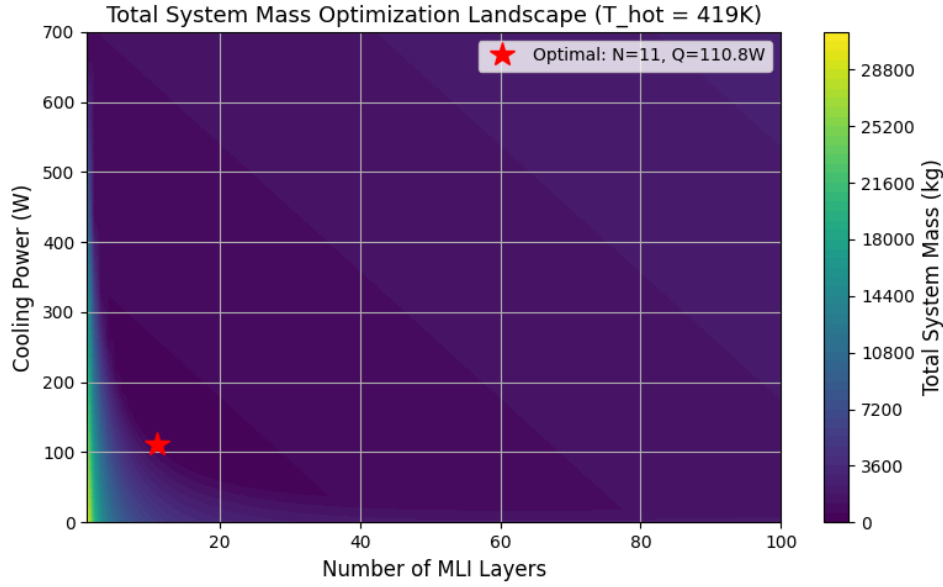


Figure 7: Total system mass optimization landscape — 2D contour plot ($T_{hot} = 419K$)

The full optimization landscape is mapped in **Figure 7**, a two-dimensional contour plot of total system mass as a function of both MLI layer count and cryocooler cooling power. The color gradient transitions from yellow (high mass) at the extremes to dark purple (low mass) in the optimal region. The contours are closely spaced near the left and bottom edges, indicating high sensitivity to insufficient insulation or insufficient cooling, while the region around the minimum is flatter, suggesting some robustness to small variations in design parameters. The global optimum, marked by a red star, coincides with 11 layers and 110 W of cooling, yielding a total mass of 350 kg, consistent with the V-curve minimum.

4.4 Mission Phase Analysis

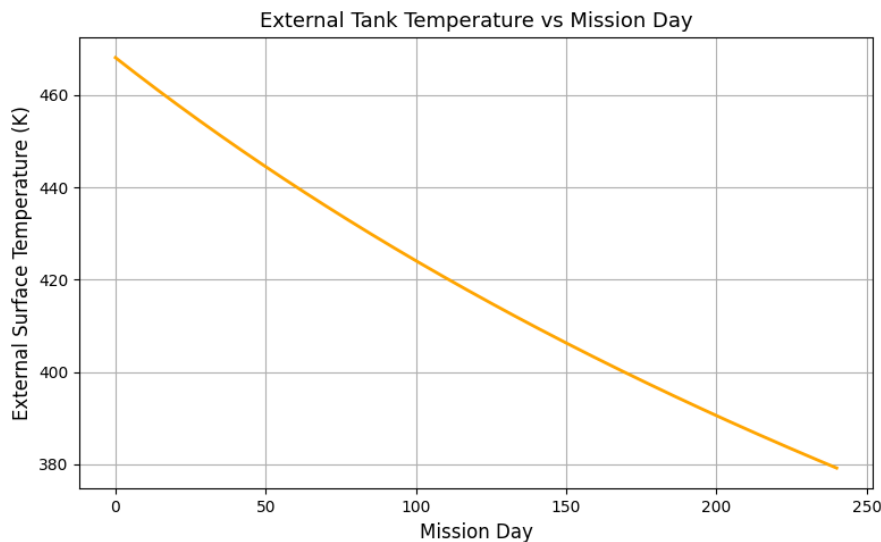


Figure 8: External tank surface temperature vs mission day over 240-day Earth-Mars transit

To incorporate the changing solar environment during the Earth–Mars transit, the hot boundary temperature was modeled as a function of distance from the Sun. **Figure 8** displays

the external layer temperature over the 240-day mission, assuming an absorptivity of 0.1 and an outer-surface emissivity of 0.05. The temperature decreases approximately linearly from 467 K at departure to 379 K at arrival, following the inverse square law of solar flux combined with the fourth-root dependence of equilibrium temperature. Even at Mars distance, the hot boundary remains well above the 300 K constant assumption often used in preliminary analyses, highlighting the need for a time-varying thermal model.

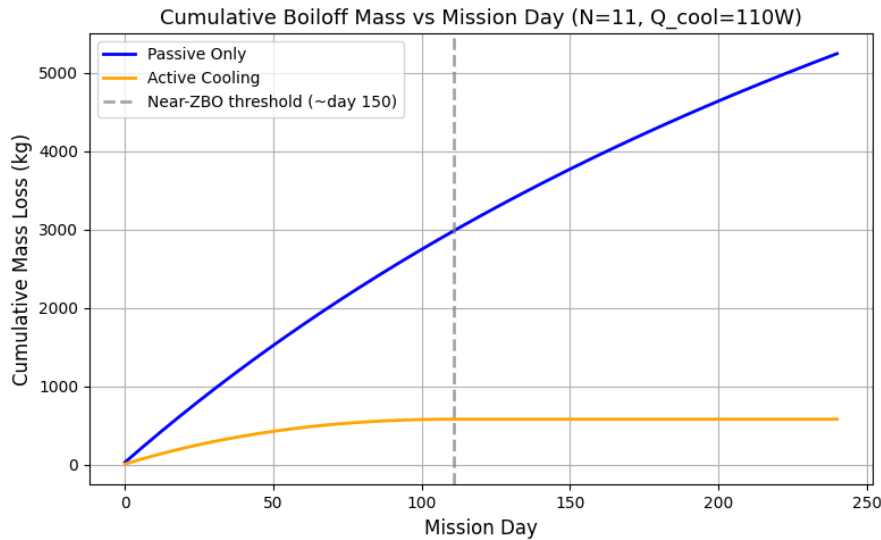


Figure 9: Cumulative boil-off mass vs mission day: passive vs active cooling ($N=11$, $Q_{cool}=110W$)

The cumulative propellant loss over the mission for both passive and active configurations at the optimized design (11 layers, 110 W cooling) is shown in **Figure 9**. The passive curve exhibits a high initial boil-off rate that gradually decelerates as the spacecraft moves away from the Sun, reaching a total loss of 5,241 kg. In contrast, the active curve flattens after approximately day 111, when the cryocooler’s 110 W cooling capacity begins to exceed the declining passive heat load, achieving near-zero daily boil-off for the remainder of the transit. The total boil-off with active cooling is 583 kg, representing a saving of 4,658 kg of LH₂ compared to the passive-only case. When contrasted with the constant-temperature assumption used in earlier parametric sweeps, which predicted only 2,683 kg of passive boil-off, the time-varying model reveals that the constant-temperature approach underestimates propellant loss by a factor of approximately two. Propellant margins must therefore account for the full thermal trajectory, particularly the elevated heat loads during the early departure phase near Earth.

5. Discussion

The hyperbolic decay in boil-off mass with increasing MLI layers, shown in Figure 1, carries a clear engineering implication: beyond approximately 30 layers, the thermal benefit of each additional layer becomes marginal while the mass penalty continues to accumulate linearly. For the 2-m sphere considered here, increasing from 30 to 60 layers reduces boil-off by only an additional 10% but doubles the insulation mass. This diminishing returns behavior defines

a practical sweet spot between 20 and 40 layers, where the trade-off between thermal performance and added mass is most favorable. Adding layers beyond this range offers diminishing thermal returns while increasing system mass, regardless of whether passive or active cooling is employed.

The most significant finding of this study is the inadequacy of constant hot-boundary temperature assumptions. The time-varying solar environment model yields an average hot temperature of 419 K over the transit, yet the cumulative boil-off cannot be captured by simply plugging this average into a steady-state model. The T_{hot}^4 dependence in the heat flux equation means that the early mission phase, when the spacecraft is near Earth and exposed to 467 K surface temperatures, dominates the total heat load. For the converged design (11 layers, 110 W cooling), the constant-temperature assumption of 300 K predicts only 2,683 kg of passive boil-off, whereas the time-varying model predicts 5,241 kg, a factor of two underestimation. This has direct implications for mission planning: propellant margins derived from average or conservative constant-temperature assumptions are insufficient. Mission designers must account for the thermal trajectory, particularly the elevated heat loads during departure and early cruise.

Active cooling provides substantial mass savings, reducing total boil-off from 5,241 kg to 583 kg at the optimized design, a reduction of nearly 90%. However, this benefit carries a cost. Achieving zero boil-off at the 11-layer design requires 110 W of cooling power, which at a COP of 0.05 demands 2,200 W of electrical input power. At Mars distance, with solar flux reduced to 590 W/m², producing this power requires approximately 3.7 m² of solar array area, translating into a power system mass of roughly 183 kg at the assumed specific mass of 0.083 kg/W. The optimization already accounts for this mass penalty, but the result is sensitive to cryocooler COP. If COP were to drop from 0.05 to 0.01, a plausible variation depending on cryocooler type and operating conditions—the required input power would quintuple to 11,000 W, making the active cooling approach impractical for a solar-powered spacecraft. This sensitivity underscores the importance of accurate cryocooler performance data [4] in early mission design.

Several limitations of the current model should be acknowledged. The trajectory is approximated as a linear increase in distance from the Sun rather than a true Hohmann transfer orbit, which would introduce variations in solar aspect angle and potential occultation periods. Only LH₂ is considered, whereas actual Mars missions would likely use both LH₂ and LOX, with different storage temperatures and thermal characteristics. The MLI mass is treated as a lumped areal density with no accounting for spacer conduction, which can contribute significantly to heat transfer at cryogenic temperatures. Tank geometry is fixed as a 2-m sphere; alternative shapes such as prolate spheroids, which offer a lower surface-area-to-volume ratio, could further reduce system mass. Structural mass of the tank itself is not included, nor is the mass of the cryocooler hardware beyond the power system penalty. Future work should incorporate higher-fidelity thermal modeling, real cryocooler performance curves, and multi-propellant analysis to reduce design uncertainty. These refinements would support a more complete mass trade-off study applicable to human-scale Mars missions.

6. Conclusion

Cryogenic propellant storage for Mars transit demands careful balancing of insulation mass, boil-off loss, and power system mass. This study developed a parametric thermal model of a spherical LH₂ tank over a 240-day Earth–Mars trajectory, incorporating a time-varying solar environment to evaluate passive MLI and active cryocooler configurations. The optimal design of 11 MLI layers with 110 W of active cooling yields a total system mass of 350 kg, reducing boil-off from 5,241 kg (passive-only) to 583 kg, a saving of 4,658 kg of propellant. Constant-temperature assumptions were shown to underestimate boil-off by a factor of two, highlighting the necessity of mission-phase thermal modeling. Future work should incorporate real cryocooler performance data, structural mass contributions, and multi-propellant analysis including LOX storage, bringing the model closer to a full preliminary design tool for human Mars missions.

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Appendix A: Python Source Code

The parametric thermal model developed in this study is implemented in Python and is publicly available at:

<https://github.com/MohammadBITS/cryogenic-tank-mars-transit>

The repository includes the complete merged analysis script, all generated figures, and a README describing the model parameters and execution instructions.

This report is also archived and citable via Zenodo:

<https://doi.org/10.5281/zenodo.19940260>

